

Self-Referential Spectral Action: Existence and Nontriviality of Fixed Points

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Problem

In the standard Connes–Chamseddine framework, the Dirac operator D_0 encodes 25+ Standard Model parameters—all externally given. We ask: **can D depend on its own spectral action value?**

Setup

Given a spectral triple $(\mathcal{A}, \mathcal{H}, D_0)$, define the spectral action

$$S[D] = \text{Tr } f(D/\Lambda) + \langle \psi, D\psi \rangle.$$

The self-referential equation is $D = D_0 + \Phi(S[D])$, $\Phi : \mathbb{R} \rightarrow \text{End}(\mathcal{H})$.

Scaling ansatz. $\Phi(S) = (\kappa \ln(S/S_0) + \beta)D_0$, where $S_0 = S[D_0]$. Set $\lambda(S) = 1 + \kappa \ln(S/S_0) + \beta$, so $D = \lambda(S)D_0$. This satisfies all spectral triple axioms for $\lambda > 0$.

Heat Kernel Expansion

Under $D = \lambda D_0$, the composite map is

$$F(S) = C_4 \lambda^{-4} + C_2 \lambda^{-2} + C_0 + \lambda E,$$

where $C_4 \sim \Lambda^4$, $C_2 \sim \Lambda^2$, $C_0 \sim \Lambda^0$. Coefficients verified against CCM06: $N = 96$ fermion degrees of freedom, $a_0 = N \cdot \text{vol}/(4\pi^2)$, the “24” factor confirmed.

Main Results

Theorem 1 (Compression). For $|\kappa| < 1/10$, $|\beta| < 0.02$, $C_4 > 0$, the map F maps $I = [0.7S_0, 1.3S_0]$ into itself with $|F'(S)| < 1$ on I . By Banach's theorem, a **unique fixed point** S^* exists.

Theorem 2 (Nontriviality). For $\beta \neq 0$, $S^* \neq S_0$. First-order correction: $\frac{S^* - S_0}{S_0} \approx -\frac{4\beta}{1 + 4\kappa}$. With $\kappa = 0.08$, $\beta = 0.01$: $S^* \approx 0.97 S_0$ (3% correction).

$\mathcal{R}$ -Invariance (Core Prediction)

The self-consistent conformal factor $\lambda^* = 1 + \kappa \ln(S^*/S_0) + \beta$ rescales all SM quantities by powers of λ^* . The dimensionless ratio

$$\mathcal{R} = \frac{\Lambda_{\text{cc}} \cdot G_N^2}{g_{\text{YM}}^4 \cdot v^4}$$

is **independent of λ^*** , hence independent of κ and β .

This is the unique robust prediction surviving the scalar bottleneck (one equation $\rightarrow$ one constraint).

Honest Limitations

- **Scalar bottleneck:** one constraint, cannot predict all 25 SM parameters.
- κ and β are **free parameters**; no physical derivation exists.
- $\Phi(S) = g(S)D_0$ is a **mathematical ansatz**, not derived from physical principles.
- **Scheme A** (inner fluctuation scaling) also works, with compression condition relaxed by $\sim \Lambda^2$.

Literature Gap

Searched arXiv (8 keyword groups), INSPIRE (8 groups), Google Scholar (5 groups). No prior work on “self-referential spectral action” or “spectral action fixed point” found. Closest: Khalkhali–Pagliaroli (arXiv:2512.08694) on bootstrapping NCG with Dirac ensembles.

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